

# Pratik Dey

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## Summary

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Distributed Systems Engineer with 4+ years of experience in cloud storage, backend infrastructure, and production reliability at Microsoft and Amazon. Particularly interested in the intersection of AI/ML systems, data infrastructure, and distributed compute including storage engines, query execution, and infrastructure for large-scale ML workloads. Portfolio: [pratik0316.github.io](https://pratik0316.github.io).

## Work Experience

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### Microsoft

Sept 2024 – Present

#### Software Engineer - Azure Storage

- Scaled **Azure Direct Drive account provisioning** throughput from **300/min** to **1000/min** through **DNS** architecture redesign.
- Eliminated subscription-level contention in **RBAC** notification handling, reducing **HTTP 409** conflicts from **150/hr** to **zero**.
- Mitigated critical security vulnerability exposing **Azure Storage SAS keys** in logs through automated detection and service hardening.
- Led migration of **4+ services** from **IFX** to **OpenTelemetry**, standardizing cross-service observability, enabling efficient in-memory metric aggregation, and reducing metric cardinality for improved telemetry scalability.
- Improved team efficiency and review consistency by building a **GitHub Copilot skill** that automates safety certification documentation for externally exposed service APIs.
- Owned production incident response for Azure Storage control plane, driving high-severity investigations, service recovery, and root cause analysis.

### Amazon

July 2022 – Aug 2024

#### Software Engineer

- Architected and scaled real-time **CDC** pipeline (**DynamoDB** → **Kinesis** → **Lambda** → **OpenSearch**) supporting **6K writes/min**, resolving hot-partitioning and consumer bottlenecks.
- Eliminated **60K+/hour** false-positive voice command triggers through redesign of command-processing logic.
- Led externalization of **Personal Knowledge Service** to Alexa mobile clients, reworking **AuthN/AuthZ** architecture and scaling backend capacity for launch readiness.
- Tech stack: Kotlin, Golang, Dagger, Guice, AWS Step Functions, ECS, Lambda, API Gateway, DynamoDB, OpenSearch, AWS Glue, CDK, React.

### CRED

Oct 2021 – Mar 2022

#### SWE Intern — SRE

- Built internal **IAM** access management portal enabling self-serve role grants across **AWS** accounts, reducing SRE on-call involvement by **50%**; implemented backend workflows using Golang, Python, Step Functions, Lambda, API Gateway, and DynamoDB.
- Built backend automation for cross-account **IAM** user migrations, handling policy mapping, permission boundaries, and rollback for seamless access transfers across ECS and Aurora-backed services.

## Achievements

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### Google Summer of Code @ AboutCode

May 2021 – Aug 2021

#### Open Source Contributor

- Enhanced the DeltaCode diff engine using VirtualCodebase, added Azure CI quality checks, and improved developer onboarding through ReadTheDocs documentation and Docker containerization.

## Technical Skills

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|-------------------------------------------|------------------------------------------------------------------------|
| Languages                                 | Python, Kotlin, Go, C#, Rust                                           |
| Cloud & Infrastructure                    | Azure, AWS, Kubernetes, Docker                                         |
| Distributed Systems & Data Infrastructure | DynamoDB, OpenSearch, Kinesis, Lambda, ECS, Apache Spark, Apache Arrow |
| Observability & Reliability               | OpenTelemetry, CI/CD, Production Debugging, Incident Response          |

## Certifications

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- Introduction to Model Context Protocol** — *Anthropic*. Covers the Model Context Protocol (MCP) standard for connecting AI models to external tools, data sources, and services through a unified interface.

## Education

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| B.E., Computer Science & Engineering | Aug 2018 – Apr 2022 |
| Jadavpur University                  | GPA: 8.83 / 10.00   |